PRITAM PATRA

SOFTWARE ENGINEERING / BACKEND SYSTEMS / FULL-STACK / APPLIED AI & ML

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GitHub / LinkedIn / Portfolio

B.Tech CSE undergraduate (8.90 CGPA) and software engineering intern at IIT Ropar, building backend services, real-time applications, and applied ML systems. Hackathon award-winner with experience shipping developer tools and technical event infrastructure.

TECHNICAL SKILLS

Languages Python, TypeScript, JavaScript, C++, SQL

Backend & systems Node.js, Express, FastAPI, Django, REST APIs, WebSockets, JWT, Docker, Linux

AI / ML Scikit-learn, PyTorch, XGBoost, LangGraph, LangChain, Pandas, NumPy, SHAP, LIME; learning Hugging Face

Frontend, data & tools React, Next.js, Tailwind CSS, PostgreSQL, MongoDB, Redis, Git/GitHub, Sentry

EXPERIENCE & LEADERSHIP

VLED Lab, IIT Ropar Software Engineering Intern (Remote) Jul 2026 - Present

- Contribute to open-source educational software through modular development, collaborative Git workflows, code reviews, and technical documentation.

MetaMorph 2.0, GNIT Head Organizer & Technical Lead Sep 2026

- Built a QR-based participant portal for unique credentials, check-in scans, present/absent attendance, food access, and room information; supported event website and venue networking.

SELECTED PROJECTS

Orbital Sentinel | GitHub FastAPI · SGP4 · XGBoost · WebSockets · CesiumJS

- Built a 3D satellite tracking and conjunction-awareness system with SGP4 propagation, 24-hour collision screening, physics-based risk estimates, and live telemetry; added an advisory XGBoost risk model.

Rawbee | GitHub React · TypeScript · Node.js · MongoDB · Cloudinary

- Engineered a video platform with JWT refresh-token auth, channels, uploads, playlists, and subscriptions; reduced measured network transfer from 5.4 MB to 471 KB (~91%).

OmniML LangGraph · PyTorch · Optuna · SHAP/LIME

- Co-developed a human-in-the-loop AutoML workflow with 13 specialized agents, editable neural-network graphs, model training, and explainability.

LOBO | GitHub MERN · Leaflet · Gemini API

- Built a local-services marketplace with provider discovery, booking workflows, map-based browsing, and an AI booking assistant.

CryptoPulse Next.js · WebSockets · PostgreSQL · Gemini · Inngest | In progress

- Developing a crypto-market dashboard with live market feeds, historical charts, watchlists, and experimental AI-assisted insights.

AWARDS

2026 Best AI/ML Track - Recursive Hackathon; Best Open Source Innovation - Aliah University Hacktoberfest; 1st Runner-Up - AKI Hacks Hacktoberfest.

EDUCATION

Guru Nanak Institute of Technology B.Tech, Computer Science & Engineering | 2024-Present | CGPA: 8.90/10